RAJJU · रज्जु · geometry by the cord

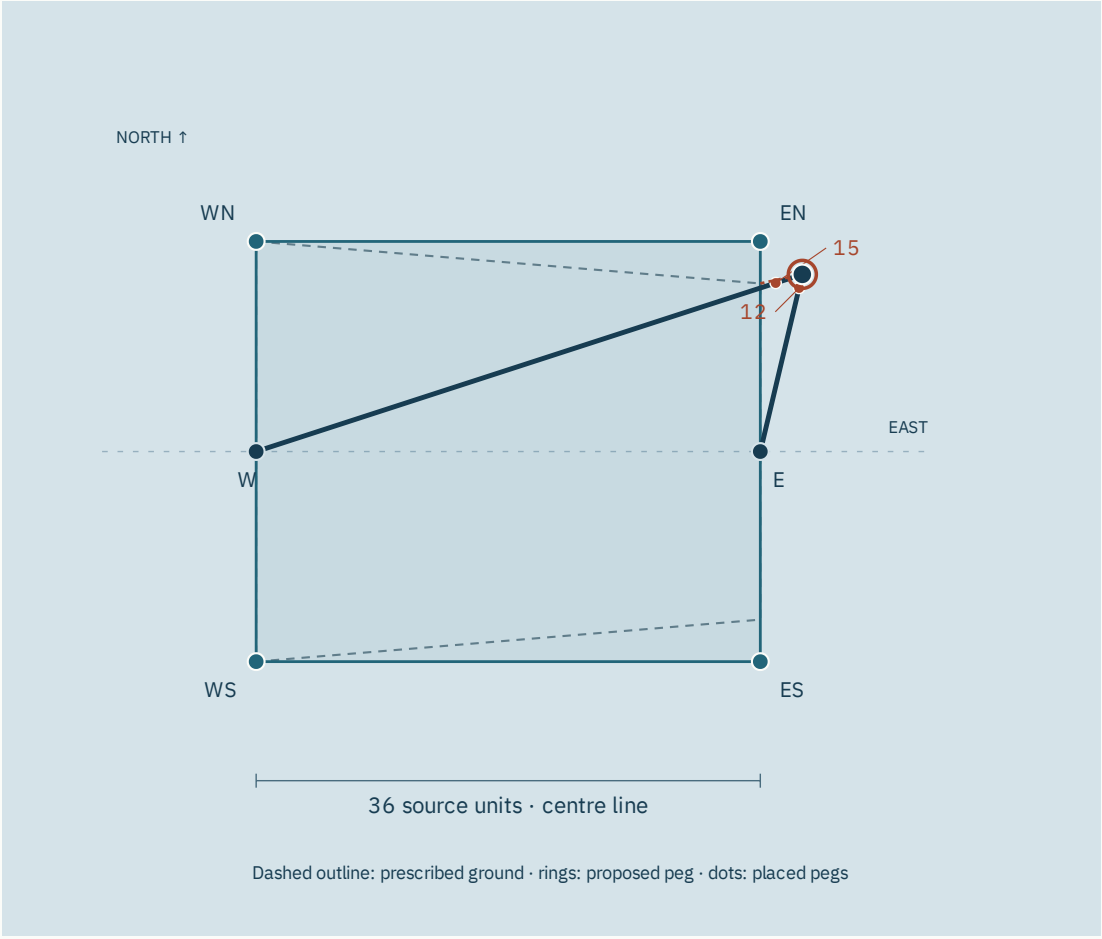
Hold fifteen. Stake twelve.

Āpastamba's ground: 36 units eastward, 30 wide at the west and 24 at the east. Source units are used consistently; no conversion to a modern physical length is inferred.

Edition bf90ece04ede9b75accf6193ec916f79ac1e1669404eaeb90758b7d7ccb0431b. Open this retained layout online. This file needs no network to read or print.

The proposed northeast peg

One cord, two marks. Fix the cord at E and 36 units inward. Cord length 54. Hold at 13; stake at 13. These distances are measured along the cord from the selected end.



The rust ring proposes a peg; the dark dot is the grip. Blue dots retain placed pegs. Dashed outline: prescribed ground.

Proposed position (39, 12.649110641); target (36, 12). Distance from target 3.069420894. Grip-to-baseline angle is 103.342363797°. The angle is measured at the centre-line end, not at a sloping corner of the final trapezoid.



Distance measured from the selected end of the cord

In the eastern one-cord operation, a grip at fifteen establishes the perpendicular. The corner peg goes at twelve on the same piece of cord. Putting the peg at the grip gives an eastern width of thirty, while the prescribed width is twenty-four.

The pegs you retained

The enclosed area is 1,080 square source units, +11.111111111% against the prescribed 972. The western width is 30; the eastern width is 30.

Corner	Method and cord marks	Position (east, north)	Distance from target
Northwest	8–15–17 and 12–35–37 Hold 15, stake 15	(0, 15)	0
Southwest	8–15–17 and 12–35–37 Hold 15, stake 15	(0, -15)	0
Northeast	One cord, two marks Hold 15, stake 15	(36, 15)	3
Southeast	One cord, two marks Hold 15, stake 15	(36, -15)	3

Changing the proposed grip does not move an already placed peg. Each placed peg retains its own method, grip and stake choice. Coordinates are recomputed from those choices.

Read the instructions

The western corners are śroṇī, haunches; the eastern ones aṃsa, shoulders. The following excerpts are from printed p. 9, with normalized Unicode, spacing and punctuation. Brackets show supplied English.

त्रिंशत्पदानि प्रक्रमावा

triṃśat padāni prakramā vā

Thirty padas or prakramas.

The western cross-line. The source names alternative measures; the experiment uses one consistent unit and makes no conversion to metres.

षट्त्रिंशत्प्राची

ṣaṭtriṃśat prācī

The eastward line [is] thirty-six.

Prācī names the eastward centre line. Its orientation is a prerequisite in the text. This experiment starts with that line already established.

चतुर्विंशतिः पुरस्तात्तिरश्ची

caturviṃśatiḥ purastāt tiraścī

The eastern cross-line [is] twenty-four.

Twenty-four is the full cross-line. The distance from the centre line to either eastern corner is twelve, not twenty-four.

द्वादशसु लक्षणं पञ्चदशसु लक्षणं

dvādaśasu lakṣaṇaṃ pañcadaśasu lakṣaṇaṃ

A mark at twelve; a mark at fifteen.

The locative endings indicate positions along the cord. Lakṣaṇa is a mark. In the eastern one-cord operation, hold at fifteen but use the twelve mark for the actual corner.

Sources and limits

This is a modern plan of Āpastamba's layout rules as excerpted and explained by Thibaut in 1875. It is not a survey of a surviving ground or a reconstruction of a complete ritual. The centre line is assumed to be oriented already. Cords are ideal, inextensible and taut; the pegs are points on a flat plane. Soil, knot thickness, cord stretch, measurement tolerances and the ritual's other requirements are outside the model. The elliptical locus is a modern mathematical comparison, not a claim about an ancient derivation.

George Thibaut, *The Śulvasūtras* (1875), pp. 9–12. Complete source volume. The printed direction discrepancy on p. 10 is retained in the page note. The optional modern ellipse and coordinate calculations do not establish the original derivation.

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Baudháyana :

त्रिकचतुष्कयोर्द्वादशिकपञ्चिकयोः पञ्चदशिकाष्टिकयोः सप्तिकचतुर्विंशिकयोर्द्वादशिकपञ्चत्रिंशिकयोः पञ्चदशिकषट्त्रिंशिकयोरित्येतास्वपलब्धिः।

This (*viz.* that the diagonal of an oblong produces by itself, &c.,) is seen in those oblongs the sides of which are three and four, twelve and five, fifteen and eight, seven and twenty-four, twelve and thirty-five, fifteen and thirty-six (literally, the sides of which consist of three parts and four parts, &c.)

This sútra contains the enumeration of, as we should say, five Pythagorean triangles, *i. e.*, rectangular triangles, the three sides of which can be expressed in integral numbers. (Baudháyana enumerates six ; but the last is essentially the same with the second, 15 and 36 being 3 × 5 and 3 × 12.) Baudháyana does not give the numbers expressing the length of the diagonals of his oblongs or the hypotenuses of the rectangular triangles, and I subjoin therefore some rules from A'pastamba, which supply this want, while they show at the same time the practical use, to which the knowledge embodied in Baudháyana's sútra could be turned.

The vedi or altar employed in the soma sacrifices was to have the dimensions specified in the following :

त्रिंशत्पदानि प्रक्रमा वा पश्चात्तिरश्ची भवति षट्त्रिंशत् प्राची चतुर्विंशतिः पुरस्तात्तिरश्चीति सौमिक्या वेदेर्विज्ञायते।

The western side is thirty padas or prakramas long, the práchí or east line (*i. e.*, the line drawn from the middle of the western side to the middle of the eastern side of the vedi) is thirty-six padas or prakramas long ; the eastern side twenty-four ; this is the tradition for the vedi at the soma sacrifices.

Now follow the rules for the measurement of the area of this vedi :

षट्त्रिंशिकायामेऽष्टादशोपसमस्यापरस्माद्दक्षादुदग्द्वादशमुत्तरं पञ्चदशमुत्तरं प्रष्ठ्याल्लयोरन्तौ नियम्य पञ्चदशिकेन दक्षिणापायम्य मध्यं निहन्त्येवमुत्तरतस्तौ यौ विपर्यस्यात्तौ पञ्चदशिकेनैवापायम्य द्वादशिकं मध्यं निहन्त्यवमुत्तरतस्तावुत्तौ तदेकरज्ज्वा विहरणम्।

Add to the length of thirty-six (*i. e.*, to a cord of the length of thirty-six either padas or prakramas) eighteen (the whole length of the cord is then 54), and make two marks on the cord, one at twelve, the other at fifteen, beginning from the western end ; tie the ends of the cord to the ends of the prishṭhyá line (the prishṭhyá is the same as the práchí, the line directed exactly towards the east and west points, and going through the centre of the vedi. The fixing of the práchí was the first thing to be done when any altar had to be measured out. The methods devised for this end will not be discussed here, as they are based on astronomical observations ; for our purpose it is sufficient to know that a line of 36 padas length

Baudhāyana's whole-number examples, followed by Āpastamba's ground dimensions and the 54-unit cord with marks at 12 and 15. The short Sanskrit readings are excerpts of this page.

and running from the east towards the west had been drawn on the ground. On both ends of this line a pole was fixed and the ends of the cord of 54 padas length tied to these poles) and taking it by the sign at fifteen, draw it towards the south; (at the place reached by the mark, after the cord has been well stretched) fix a pole. Do the same on the northern side (*i. e.*, draw the cord towards the north as you have drawn it just now towards the south). By this process the two s'roṇís, the southwest corner and the southeast corner of the vedi are fixed. After that exchange (the ends of the cord; *i. e.*, tie that end which had been fastened at the pole on the east end of the práchí to the pole on its west end and *vice versâ*), and fix the two aṃsas ("shoulders" of the vedi, *i. e.*, the southeast corner and the northeast corner). This is done by stretching the cord towards the south having taken it by the mark at fifteen and by fixing a pole on the spot reached by the mark at twelve; and by repeating the same operation on the northern side. The result are the two aṃsas. This is the measurement of the vedi by means of one cord (the measurements described further on require two cords each). (See diagram 1.)

The whole process described in the preceding is founded on the knowledge that a triangle, the three sides of which are equal to 15, 36, 39, is rectangular.

The end aimed at was to draw the east and the west side of the vedi at right angles on the práchí. Accordingly, the práchí a b being 36 feet long, a cord a c b (= 54) was divided by a mark into two parts a c = 39 and b c = 15 and fastened at a and b. If then this cord was taken at c, and stretched towards the right, the angle a b c could not but be a right angle. The same applies to the angles a b d, b a e, and b a f. In fixing the two east corners, both marks on the cord had to be employed, the mark at fifteen being used for constructing the right angle, the mark at 12 giving to the east side of the vedi the prescribed length (24 padas).

चिकचतुष्कयोः पञ्चिकाक्ष्णयारज्जुः ।

The diagonal cord of an oblong, the side cords of which are three and four, is five.

ताभिस्त्रिरभ्यस्ताभिरक्ष्णौ ।

With these cords increased three times (by itself; *i. e.*, multiplied by four) the two eastern corners of the vedi are fixed.

The proceeding is as follows: (See diagram 2.)

At c, at a distance of 16 padas from a, the east end of the práchí, a pole is fixed and then a cord of 32 feet length tied to the poles at a and c. The cord is marked at a distance of 12 padas from a, and then taken by the mark and drawn towards the south until it reaches the position a e c. Thus

Continue the one-cord procedure: reverse the ends, hold at 15, stake at 12. The English prints southwest and southeast for the first pair; the described north/south operation at the western end gives northwest and southwest in the modern plan. The 3–4–5 method starts below.

a triangle is formed, the sides of which are 12, 16, 20 and this triangle is a rectangular one; a e stands at right angles on a c, and as it is just 12 padas long, e marks the place of the southeast corner of the vedi. The north east corner d is found in the same way.

चतुरभ्यक्षाभिः श्रोणी ।

With the same cords increased four times (*i. e.*, their length multiplied by five) the two western corners of the vedi are found.

In this case a cord of 40 padas length is tied to the poles at c and b, and marked at the distance of 15 padas from b. Then it is taken by the mark and drawn towards the south into the position b g c. The result is a rectangular triangle as above; g marks the place of the southwest corner. The same operation repeated on the north side gives f as the place of the northwest corner of the vedi.

Another method for the measurement of the vedi follows:

द्वादशिकपञ्चिकयोरक्षयोर्दशिकाक्ष्णयारज्जुरक्षाभिरंसौ ।

The diagonal cord of an oblong, the sides of which are twelve and five, is thirteen; with these cords the two east corners are fixed.

(See diagram III.)

A pole is fixed at the distance of five padas from the east end of the práchí, a cord of twenty-five padas length fastened at a and c, marked at the distance of 12 padas from a, drawn towards the south &c., as above.

त्रिरभ्यक्षाभिः श्रोणी ।

With these cords increased twice (multiplied by three) the two western corners are fixed.

The requisite rectangular triangle is here formed by the whole práchí = 36, and by a cord of 54, divided by a mark into two pieces of 15 and 39.

Another method follows:

पञ्चदशिकाष्टिकयोः सप्तदशिकाक्ष्णयारज्जुरक्षाभिः श्रोणी ।

The diagonal cord of an oblong, the sides of which are fifteen and eight, is seventeen; with these cords the two western corners are fixed.

(See diagram 4.)

A pole b is fixed at the distance of eight padas from d, a cord of 32 padas tied to b and d, &c.

द्वादशिकपञ्चत्रिंशिकयोः सप्तत्रिंशिकाक्ष्णयारज्जुरक्षाभिरंसौ ।

The diagonal cord of an oblong, the sides of which are twelve and thirty-five is thirty-seven; with these cords the two eastern corners are fixed.

A pole is fixed at c, thirty-five padas to the west from a; a cord of forty-nine padas tied to a and c, &c.

Continuation of the enlarged 3–4–5 method, then 5–12–13 and the paired 8–15–17 / 12–35–37 constructions. The modern plan derives each auxiliary peg from the stated distance inward along the centre line.

एतावन्ति विज्ञेयानि वेदिविहरणानि भवन्ति ।

So many "cognizable" measurements of the vedi exist.

That means: these are the measurements of the vedi effected by oblongs, of which the sides and the diagonal can be known, *i. e.*, can be expressed in integral numbers.

In this manner A'pastamba turns the Pythagorean triangles known to him to practical use (the fourth of those which Baudháyana enumerates is not mentioned, very likely because it was not quite convenient for the measurement of the vedi), but after all Baudháyana's way of mentioning these triangles as proving his proposition about the diagonal of an oblong is more judicious. It was no practical want which could have given the impulse to such a research—for right angles could be drawn as soon as one of the "vijneya" oblongs (for instance that of 3, 4, 5) was known—but the want of some proof which might establish a firm conviction of the truth of the proposition.

The way in which the Sútrakáras found the cases enumerated above, must of course be imagined as a very primitive one. Nothing in the sútras would justify the assumption that they were expert in long calculations. Most likely they discovered that the square on the diagonal of an oblong, the sides of which were equal to three and four, could be divided into twenty-five small squares, sixteen of which composed the square on the longer side of the oblong, and nine of which formed the area of the square on the shorter side. Or, if we suppose a more convenient mode of trying, they might have found that twenty-five pebbles or seeds, which could be arranged in one square, could likewise be arranged in two squares of sixteen and of nine. Going on in that way they would form larger squares, always trying if the pebbles forming one of these squares could not as well be arranged in two smaller squares. So they would form a square of 36, of 49, of 64, &c. Arriving at the square formed by $13 \times 13 = 169$ pebbles, they would find that 169 pebbles could be formed in two squares, one of 144 the other of 25. Further on 625 pebbles could again be arranged in two squares of 576 and 49, and so on. The whole thing required only time and patience, and after all the number of cases which they found is only a small one.

Having found that, in certain cases at least, it was possible to express the sides and the diagonal of an oblong in numbers, the Sútrakáras naturally asked themselves if it would not be possible to do the same thing for a square. As the side and the diagonal of a square are in reality incommensurable quantities we can of course only expect an approximative value; but their approximation is a remarkably close one.

Baudháyana:

प्रमाणं तृतीयेन वर्धयेत्तच्च चतुर्थेनात्मचतुस्त्रिꣳशोनेन । सविशेषः ।

Closing explanation of known whole-number measurements, then the beginning of a different numerical-diagonal discussion. The editor's proposed discovery history is not required by the geometric model.